

ERIE, PA, USA

WMO: 725260

Lat: 42.080N

Lon: 80.182W

Elev: 222

StdP: 98.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14860

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-12.4	-20.0	0.7	-14.3	-17.4	0.8	-11.4	12.3	3.2	11.6	0.5	4.4	190	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	30.5	22.8	29.1	22.1	27.9	21.4	24.0	28.4	23.2	27.4	22.5	26.6	4.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	17.8	27.0	21.8	16.9	26.1	21.0	16.1	25.4	73.2	28.6	69.9	27.5	67.1	26.6	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.4	8.6	-17.6	33.1	4.6	1.0	-20.9	33.9	-23.6	34.5	-26.2	35.0	-29.6	35.8		
(5)			-18.2	25.4	4.4	0.8	-21.3	26.0	-23.9	26.5	-26.3	26.9	-29.5	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-2.6	-1.9	2.0	8.3	14.6	19.9	22.3	21.7	18.3	12.3	6.2	1.0		
	DBStd	10.10	6.26	6.23	6.10	5.27	4.93	3.87	2.94	2.79	3.85	4.52	4.90	5.04		
	HDD10.0	1541	391	337	260	96	12	0	0	0	1	26	136	283		
	HDD18.3	3345	647	568	507	304	138	27	3	4	47	196	364	538		
	CDD10.0	1625	2	3	12	44	154	297	381	363	249	96	21	3		
	CDD18.3	389	0	0	1	3	23	74	126	109	46	8	0	0		
	CDH23.3	2532	0	0	3	27	159	504	882	652	271	34	0	0		
	CDH26.7	551	0	0	0	3	29	125	201	131	58	4	0	0		
(14) Wind	WSAvg	4.3	5.2	4.7	4.6	4.4	3.9	3.6	3.4	3.4	3.8	4.4	4.9	5.0		
(15) Precipitation	PrecAvg	1095	70	61	75	89	88	103	91	102	116	97	105	98		
	PrecMax	1628	136	147	172	196	162	222	228	300	270	210	264	176		
	PrecMin	791	25	16	17	41	25	19	13	13	34	34	39	42		
	PrecStd	161	28	31	32	36	36	48	43	56	59	36	44	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	18.0	22.6	26.6	29.6	31.5	32.1	31.5	31.0	26.7	20.7	17.7		
		MCWB	13.6	13.0	15.3	17.3	21.2	22.8	23.6	23.4	22.7	19.8	15.0	13.6		
	2%	DB	13.3	13.4	18.3	23.1	27.1	29.7	30.3	29.7	28.3	23.8	18.1	13.6		
		MCWB	10.9	10.0	12.6	15.3	19.7	22.0	23.1	22.7	21.5	17.9	13.4	10.2		
	5%	DB	9.4	9.8	15.0	20.1	25.1	28.1	28.9	28.1	26.3	21.5	16.1	10.8		
		MCWB	6.7	6.4	10.7	13.4	18.3	21.2	22.2	22.0	20.5	16.8	12.3	8.5		
	10%	DB	6.1	6.4	11.5	17.2	22.8	26.4	27.5	26.7	24.6	19.2	13.9	7.9		
		MCWB	3.9	3.8	7.5	11.9	17.2	20.5	21.6	21.3	19.7	15.0	10.2	5.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	13.9	15.8	18.4	22.2	24.5	25.3	25.0	24.1	21.2	16.3	14.1		
		MCDB	16.2	17.6	21.3	24.7	27.9	29.2	30.2	29.5	29.2	25.3	19.1	16.5		
	2%	WB	11.1	10.1	13.6	16.3	20.6	23.0	24.1	23.9	22.6	19.3	14.4	11.2		
		MCDB	13.1	13.4	17.2	21.7	25.9	28.1	28.6	27.7	26.7	22.6	17.2	13.3		
	5%	WB	7.1	6.7	10.8	14.5	19.3	22.1	23.3	23.0	21.4	17.4	12.6	8.2		
		MCDB	9.2	9.2	14.4	19.1	23.9	26.9	27.4	26.6	25.3	20.6	15.6	10.4		
	10%	WB	4.0	3.6	7.7	12.4	17.9	21.1	22.5	22.1	20.3	15.6	10.6	5.7		
		MCDB	5.9	6.1	11.1	16.8	22.1	25.3	26.5	25.8	23.8	18.7	13.5	7.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.4	7.9	9.3	9.2	8.5	8.1	8.0	8.2	7.6	6.6	5.8		
		MCDBR	10.3	11.6	12.8	13.9	11.3	10.0	9.3	9.3	9.8	10.1	9.8	9.5		
		MCWBR	8.9	9.2	8.6	8.5	6.3	5.3	5.0	5.2	5.4	6.0	7.3	8.1		
		MCWBR	10.1	11.1	12.1	12.9	10.4	8.8	7.8	7.9	8.5	8.7	9.2	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.341	0.343	0.380	0.414	0.440	0.438	0.424	0.384	0.361	0.316	0.299		
		taud	2.406	2.245	2.330	2.319	2.263	2.230	2.261	2.291	2.373	2.421	2.519	2.496		
	Ebn at Noon		841	851	901	889	863	838	835	836	848	825	826	820		
		Edh at Noon	79	109	113	123	133	138	133	125	107	91	71	67		
(44) All-Sky Solar Radiation	RadAvg		1.42	2.15	3.25	4.46	5.43	5.98	6.00	5.31	4.20	2.62	1.65	1.02		
	RadStd		0.11	0.26	0.39	0.47	0.52	0.45	0.44	0.28	0.40	0.26	0.24	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	N/A

Nomenclature: See separate page